



Managing Infection Following Scoliosis Correction

Maria Anastasia

Impact on Patient Recovery



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graph TD; A[Impact on Patient Recovery] --> B[Risk of Complications]; B --> C[Healthcare Costs]; C --> D[Patient Satisfaction and Quality of Life]
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Risk of Complications

Healthcare Costs

Patient Satisfaction and Quality of Life

The Importance of Managing Infections Post-scoliosis Correction Surgeries

Why Scoliosis Correction Surgery Poses High-Risk of SSI?

Original Article

Reducing Surgical Site Infection in Pediatric Scoliosis Surgery: A Multidisciplinary Improvement Program and Prospective 4-Year Audit

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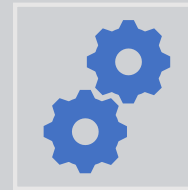
Global Spine Journal
2020, Vol. 10(5) 633-639
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DOI: 10.1177/2192568219868200
journals.sagepub.com/home/gsj
SAGE



Lengthy procedure



Extensive exposure



Use of
instrumentation

Treatment Approach

Early diagnosis and assessment

- Imaging and laboratory studies

Antibiotic therapy

- Steinberg et al. → infection rates were lower with antibiotic administration within 30 minutes of incision

Surgical intervention

- Wound exploration and debridement

Advanced wound care technique

- Negative pressure wound therapy (NPWT)

Case 1



1st presentation
(One year after scoliosis surgery)



2nd presentation: Pus containing lump
(12 days later)



1st debridement



**3 months after
debridement**



**Post- 2nd
debridement
culture**



1st VAC application



**4 weeks after 1st
VAC application**



**2 weeks after 2nd
VAC application**



**3rd application of
VAC
(4 weeks later)**

Case 2



1st presentation
(21 days after surgery)



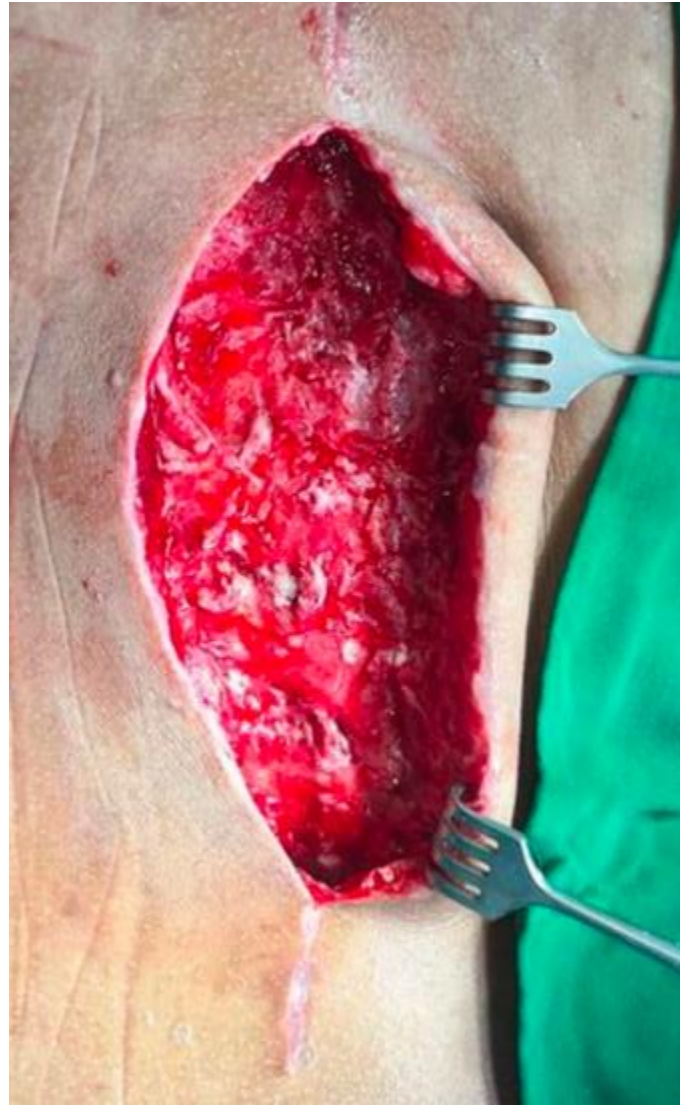
2nd presentation
(5 days later)



3rd presentation
(4 days later/4 weeks after surgery)



Sample culture



Debridement and wound exploration
(4 weeks after surgery)



8 weeks after debridement

Case 3



1st presentation
(10 days after surgery)



2nd presentation
(3 days later)



Wound debridement & exploration



14 days after debridement



4 weeks after debridement



8 weeks after debridement

Case 4



1st presentation
(11 days after surgery)



**Wound debridement &
exploration**
(14 days after surgery)



Wound closure



7 days after
debridement



14 days after
debridement



3 weeks after
debridement



VAC after 2nd
debridement



30 days after 2nd
debridement

What were the causative pathogen?

Case 1: *Enterobacter cloacae*

Gram negative bacteria → Nosocomial pathogen

Case 2: *Klebsiella pneumoniae*

Gram negative bacteria → Common cause of SSI

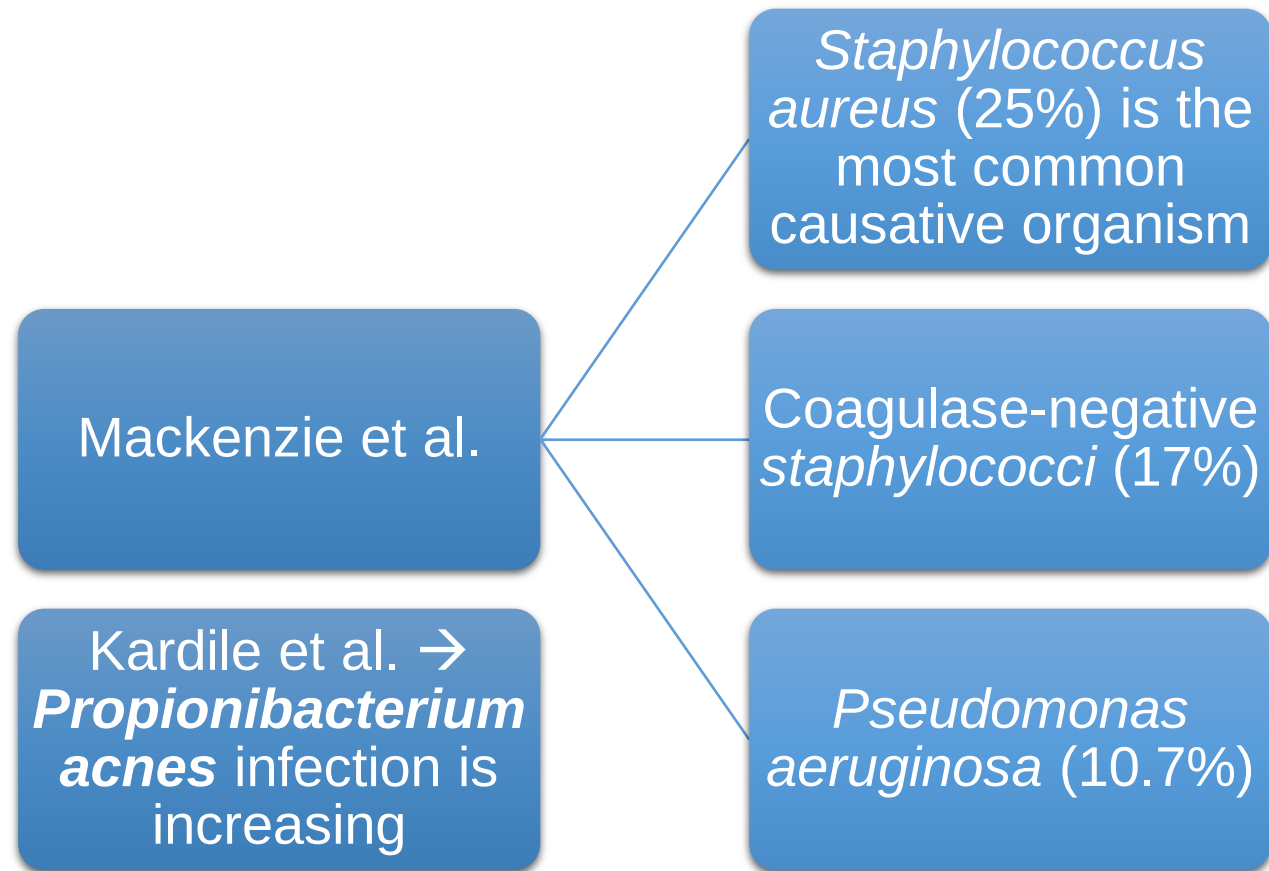
Case 3: *Escherichia coli*

Gram negative bacteria → Common cause of SSI

Case 4: *Acinetobacter baumannii*

Gram negative bacteria → Known for its high resistance to antibiotics

The most common organisms involved:



SCIENTIFIC ARTICLES

Surgical Site Infection Following Spinal Instrumentation for Scoliosis

A Multicenter Analysis of Rates, Risk Factors, and Pathogens

Mackenzie, W.G. Stuart BS, MA¹; Matsumoto, Hiroko MA¹; Williams, Brendan A. BA¹; Corona, Jacqueline MD²; Lee, Christopher MD³; Cody, Stephanie R. BS⁴; Covington, Lisa RN, MPH¹; Saiman, Lisa MD, MPH¹; Flynn, John M. MD⁴; Skaggs, David L. MD³; Royce, David P. Jr. MD¹; Vitale, Michael G. MD, MPH¹

Author Information

The Journal of Bone & Joint Surgery 95(9):p 800-806, May 1, 2013. | DOI: 10.2106/JBJS.L.00010

Spine Deformity (2021) 9:757–767
<https://doi.org/10.1007/s43390-020-00250-x>

CASE SERIES

Is *Propionibacterium acnes* becoming the most common bacteria in delayed infections following adolescent idiopathic scoliosis surgery?

Mayur P. Kardile¹ · Sukhraj S. Bains² · Calvin C. Kuo¹ · Todd L. Lincoln¹ · Ravi S. Bains¹



What were the causative pathogen?

Case 1: *Enterobacter cloacae*

Gram negative bacteria → Nosocomial pathogen

Case 2: *Klebsiella pneumoniae*

Common in:

- Procedures > 2 hours
- Large surgical sites
- Use of implants

Case 3: *Escherichia coli*

- Gut flora → Usually contaminate the surgical site during post-surgical care

Case 4: *Pseudomonas aeruginosa*

- Common cause of post-deformity correction SSI
- Nosocomial pathogen

Nosocomial Infections in Post-Correction Deformity Surgeries: Still a Challenge



Hot, humid climate may be optimal for proliferation of bacteria



Hospital Environment: Multiple pathogens thrive in healthcare settings



Multi-bed ward: Multiple patients in close-proximity raise the risk of pathogen spread



3 out of 4 cases have shown the efficacy of vacuum dressing in managing patients with post-deformity correction SSI

Is vacuum-assisted closure (VAC) system important in managing post-deformity correction SSI?



The Spine Journal 7 (2007) 596–600



Vacuum-assisted wound closure of deep infections after instrumented spinal fusion in six children with neuromuscular scoliosis

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ORIGINAL ARTICLE

Postoperative Deep Wound Infections in Adults After Spinal Fusion Management with Vacuum-Assisted Wound Closure

Mehbod, A.A. MD¹; Ogilvie, J.W. MD²; Pinto, M.R. MD³; Schwender, J.D. MD⁴; Transfeldt, E.E. MD⁵; Wood, K.B. MD⁶; Le Huec, J.C. MD⁷; Dressel, T. MD⁸

[Author information](#)

Journal of Spinal Disorders & Techniques 18(1):p 14–17, February 2005. | DOI: 10.1097/01.bsd.0000133493.32503.d3



Vacuum-assisted closure for deep infection after spinal instrumentation for scoliosis

1. Canavese F, Gupta S, Krajchich JJ, Emara KM. Vacuum-assisted closure for deep infection after spinal instrumentation for scoliosis. *J Bone Joint Surg Br.* 2008;90-B(3):377–381. doi:10.1302/0301-620X.90B3.19890
2. Mehbod, A.A. et al. Postoperative Deep Wound Infections in Adults After Spinal Fusion: Management with Vacuum-Assisted Wound Closure. *Journal of Spinal Disorders & Techniques* 18(1):p 14–17, February 2005. | DOI: 10.1097/01.bsd.0000133493.32503.d3
3. van Rhee, M. A., de Klerk, L. W. L., & Verhaar, J. A. N. (2007). Vacuum-assisted wound closure of deep infections after instrumented spinal fusion in six children with neuromuscular scoliosis. *The Spine Journal*, 7(5), 596–600. <https://doi.org/https://doi.org/10.1016/j.spinee.2006.09.002>

TAKE HOME MESSAGES

- **Surgical Technique:** Rigorous sterile procedures during surgery minimize contamination.
- **Antibiotic Prophylaxis:** Administer prophylactic antibiotics based on local resistance patterns.
- **Wound Care:** Maintain a clean and dry incision site to prevent infection.
- **Patient Hygiene:** Promote proper patient hygiene practices
- **Comorbidity Management:** Manage underlying health conditions that weaken the immune system.
- **Operating Room Environment:** Utilize a negative pressure operating room to control airborne contaminants.
- **Post-operative Monitoring:** Closely monitor patients for signs and symptoms of infection.
- **Multidisciplinary Collaboration:** Foster collaboration among healthcare professionals to prevent SSIs.



THANKYOU

